

Cera One sEMG Accessibility Band

Technical Overview and Prototype Development

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1 A wearable interface for accessible control

Cera One is a wearable interface in development that will translate subtle muscle activity into smartphone and connected-device commands. Using surface electromyography (sEMG), the band will capture activity at the wrist or lower forearm and transmit it over Bluetooth. A phone or another compatible Bluetooth-connected host will classify gestures and turn recognized intent into an action.

Prototype work to date includes an early PCB layout and a wearable form-factor visualization. The next integrated hardware design centers on two bipolar sensing channels, a shared bias contact, an ADS1292R analog front end, and an nRF52840 controller. The band will handle acquisition and wireless transmission, with lightweight onboard signal processing anticipated. Gesture classification will run on the connected host, not on the band.



Figure 1. Cera One P02 form-factor visualization, showing a woven strap, paired contact modules, a translucent electronics enclosure, and an external button. The image presents the wearable design direction rather than validated internal hardware or measured performance.

A focused first interaction set. Pinch will map to play/pause, fist to volume down, and wrist extension to volume up. Rest and unrecognized activity will produce no action. Personal calibration will adapt recognition to the wearer; additional gestures will follow only when they can be recognized reliably and performed comfortably.

Development targets. The initial system targets at least 90% held-out accuracy for one wearer, fewer than one false command per 30 minutes of rest, median end-to-end latency below 200 ms, a 3 m wireless link, at least four hours of runtime, and a hardware cost below \$150. These are performance goals; integrated-system validation remains ahead.

2 Architecture: sensing on the band, intelligence on the host

A clear division of responsibility

Cera One separates the body-worn sensing hardware from gesture recognition. The band will acquire synchronized muscle signals and deliver them to a compatible host. The phone or connected device will run the personalized classifier, reject ambiguous activity, and execute supported commands. This architecture will allow recognition software and gesture mappings to evolve without replacing the wearable electronics.

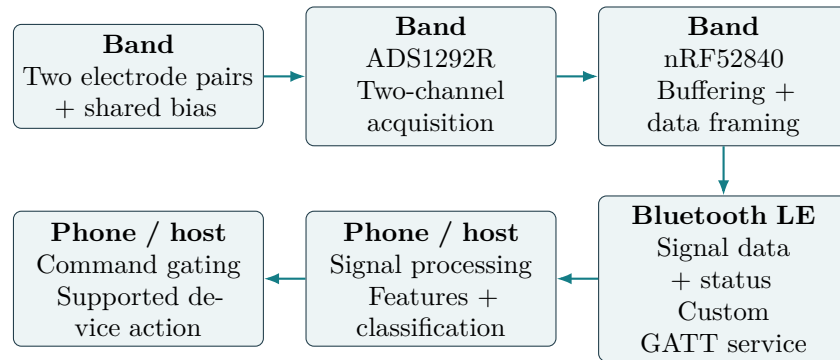


Figure 2. Target system architecture. Configuration and acquisition control will also travel from the host back to the band. Some preprocessing is expected to move onto the band as firmware develops.

Two-channel sensing hardware

Two bipolar electrode pairs and one shared bias contact will provide five skin contacts. The ADS1292R design baseline provides two simultaneous 24-bit channels with programmable gain and a serial peripheral interface (SPI). The nRF52840 will receive both channels at a target rate of 1 kS/s per channel and prepare the stream for Bluetooth Low Energy (BLE) transmission.

The two-channel approach focuses the first integrated design on repeatable acquisition rather than a large sensor array. Electrode placement will be personalized across candidate flexor and extensor regions. Input protection, bias circuitry, supply regulation, and analog filtering will be addressed in the revised electrical design and verified before body-connected use.

Processing and data integrity

Lightweight preprocessing on the band is an anticipated part of the architecture; the exact allocation of filtering and signal-quality checks will be finalized through implementation measurements. Classification and command decisions will remain host-side. The stream will identify its sample rate, channel order, and preprocessing configuration so the host can interpret each recording consistently.

Acquisition buffering will be independent of Bluetooth delivery. Sequence numbers and acquisition timing will identify missing or stale data, and the host will suppress commands whenever the received signal is invalid. A live sensing stream will be required for normal operation; the band will not act as a standalone gesture classifier when disconnected.

3 Prototype development and wearable design

Early PCB development

The early PCB layout documents the project's first electronics design work. It includes an ESP32-C6-DevKitM-1 module, five repeated INA128-labeled amplifier areas, 47 μF capacitors, a battery connector, and an L7905 footprint. Routed copper and defined antenna and USB regions capture the initial approach to analog acquisition, processing hardware, and portable power.

This layout is the record of the earlier hardware iteration, not the ADS1292R/nRF52840 revision. The integrated wearable will move to two sensing channels and a revised power and protection design. The distinction preserves the work already completed while making the direction of the next hardware revision explicit; fabrication and electrical verification are not established by the layout image alone.

Design area	Work represented to date	Integrated wearable direction
Electronics	Early ESP32-C6 and INA128 PCB layout.	Two-channel ADS1292R acquisition with an nRF52840 wireless controller.
Recognition	A focused gesture vocabulary and personalized recognition baseline.	Classification and command gating on the phone or compatible connected host.
Wearable form	P02 visualization of the strap, enclosure, and paired contact modules.	Adjustable fitting, protected electronics, and repeatable electrode positioning.
Power and access	Battery, antenna, and USB regions in the earlier layout.	Battery-only wear, revised power circuitry, and wired access outside wear sessions.

Form factor and electrode integration

The P02 visual design brings the electronics and skin-contact system into a single band. Its woven strap and modular contact arrangement establish a direction for adjustable fitting, while the enclosed electronics and external button provide a basis for protected packaging and local control. The button's intended role will include disabling command output.

Initial signal characterization will use disposable electrodes to establish a consistent sensing baseline. Integrated contacts will be evaluated against that baseline as the wearable develops. Contact material, spacing, the shared bias-contact location, and enclosure dimensions remain part of hardware refinement rather than specifications established by the image.

Repeatable fit and serviceable construction

The mechanical design will prioritize stable contact, low strain on the electrode interconnects, and accessible fitting. Placement records will link each calibration profile to the band orientation and anatomical landmarks. Refit testing will quantify how recognition changes after the band is removed and replaced.

The enclosure will retain and protect the battery while allowing assembly and maintenance. Packaging development will balance size and comfort with antenna clearance, lead routing, and reliable contact during ordinary arm movement. These mechanical decisions are integral to signal quality, not simply the appearance of the finished band.

4 Signal processing and host-side recognition

A personalized recognition pipeline

The first recognition baseline will use 1 kS/s per channel and a 20–450 Hz processing band, with a selectable 50 or 60 Hz notch. Filtering will be coordinated across the band and host; the initial implementation will establish which lightweight operations run on the wearable. Feature extraction and classification will run on the host in the baseline architecture.

The host will evaluate 200 ms windows every 50 ms, corresponding to 200 samples per channel and a 50-sample hop. Each channel will contribute five time-domain features: mean absolute value (MAV), root mean square (RMS), waveform length (WL), zero crossings (ZC), and slope sign changes (SSC). For a window of N samples, the amplitude and length features are

$$\text{MAV} = \frac{1}{N} \sum_{i=1}^N |x_i|, \quad \text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}, \quad \text{WL} = \sum_{i=2}^N |x_i - x_{i-1}|.$$

Noise thresholds will limit spurious ZC and SSC counts. The resulting ten-feature representation will feed a personalized linear discriminant analysis (LDA) classifier. Normalization and classifier parameters will be fitted from the wearer’s training recordings, with separate trials reserved for tuning and evaluation.

From a prediction to an intentional command

Recognition will be followed by a host-side decision gate. Its role is to turn sustained, valid intent into a single action rather than treating every predicted window as a new command.

1. **Signal validity.** Missing samples, stale data, clipping, and excessive artifact will suppress output.
2. **Activation and confidence.** A gesture will need to satisfy wearer-specific activation and classification thresholds. Rest and unknown activity will both produce no command and remain distinct in diagnostics.
3. **Confirmation.** A persistence rule across consecutive updates will reduce transient detections. Its duration will be tuned against responsiveness and false-command measurements.
4. **Single-action output.** Each accepted activation will produce one mapped action. A held gesture will not repeatedly trigger play/pause or volume changes in the initial interface.
5. **Release and re-arming.** A return to rest and a refractory interval will precede the next action. Faults and reconnections will clear pending decisions and require a fresh activation.

Responsiveness across the complete system

The latency target covers muscle onset through the visible or audible response, including Bluetooth transport, host processing, and command execution. A rolling 200 ms window can include pre-onset samples, so its duration is not automatically a fixed 200 ms delay. Nevertheless, window alignment, filtering, confirmation, and host scheduling all affect response time. Measurements will determine the final window and gating settings; the below-200 ms median remains a target, not a demonstrated result.

5 Connected-device experience and Bluetooth integration

The host as the interaction hub

The Flutter companion app will provide the phone-based experience: fitting guidance, signal diagnostics, personalized calibration, gesture recognition, and supported command mappings. Another compatible Bluetooth-connected device will be able to serve as the host when it implements the sensing protocol, recognition pipeline, and action interface. Bluetooth connectivity alone will not establish that compatibility.

A custom BLE Generic Attribute Profile (GATT) service will carry signal data and device status from the band. Configuration and acquisition control will return over the same connection. Gesture identifiers, classification scores, and command events will originate on the host, rather than arrive as completed classifications from the wearable.

System layer	Information or function	Role in the experience
Band-to-host stream	Synchronized channel data, sequence numbers, timing, and quality/status information.	Supplies the live input needed for host-side recognition.
Host-to-band control	Acquisition settings, streaming control, and preprocessing configuration.	Keeps the sensing configuration aligned with the calibration profile.
Host recognition	Features, gesture predictions, confidence scores, and command gating.	Converts signal data into accepted actions while rejecting ambiguous activity.
Host action interface	Platform-supported media and accessibility controls.	Executes each accepted mapping once and exposes only supported capabilities.

Continuous sensing, controlled output

At the baseline sample rate, two packed 24-bit channels represent $2 \times 1,000 \times 3 = 6,000$ bytes per second of sample payload, before timestamps, framing, or Bluetooth overhead. Transport evaluation will include sustained delivery and packet batching, because excessive buffering can add latency even when average throughput is sufficient.

Pairing and an active, valid stream will be required before command output is enabled. Disconnects will disable output and clear pending actions; reconnection will establish a new sensing session without replaying old commands. The host will execute media actions through its supported platform interfaces. Direct band-generated HID media commands are not the baseline control path in this host-classification architecture.

Personal calibration and accessible setup

Setup will guide the wearer through fitting, a rest recording, and repeated comfortable examples of each gesture. The host will build a personal profile and verify it on separate trials before enabling interactive use. Profiles will retain placement notes, processing versions, and thresholds. The interface will make connection and output-enabled states explicit and provide breaks, assistance options, and a clear stop control. Available mappings will reflect the connected platform’s permissions and capabilities.

6 Performance objectives and evaluation

Cera One’s evaluation will measure both recognition quality and the reliability of the complete band-to-host interaction. A high classifier score alone will not establish a usable interface: unintended commands, response time, fitting consistency, and comfort will be evaluated alongside recognition accuracy. The figures below are development targets, not completed test results.

Evaluation area	Evidence to be reported	Target or criterion
Electrical acquisition	Supply behavior, gain, timing, noise, clipping, and recovery under isolated bench tests.	Reviewed electrical design and verified operation before supervised skin contact.
Gesture recognition	Whole-trial held-out evaluation, confusion matrix, and per-class precision and recall.	At least 90% held-out accuracy for one wearer, with class balance and scoring unit disclosed.
Unintended commands	Emitted actions during quiet rest and representative non-command movement.	Fewer than one false command per 30 minutes of rest; zero events in the initial 30-minute screen.
Response time	Muscle-onset-to-action timing, including transport and host execution.	Median below 200 ms; 95th percentile and missed actions also reported.
Wireless operation	Sustained signal delivery, 3 m range, disconnects, and reconnects.	No duplicate or replayed command; invalid or stale data suppresses output.
Refit and usability	Recognition after removal and replacement, calibration burden, and wearer feedback.	Documented performance change, recalibration need, and comfort findings.
Battery operation	Runtime and current consumption with continuous sensing and transmission.	At least four hours in the intended operating mode.
Phone compatibility	Supported actions on tested Android and iOS configurations.	Device, OS, app state, permissions, and unavailable mappings disclosed.

Evaluation that reflects actual use

Training and test sets will be separated by whole trials or sessions. Overlapping windows from the same contraction will remain in the same split, preventing near-identical signal segments from inflating performance. Window-level recognition and command-level behavior will be reported separately, including the effect of host-side rejection and gating.

A zero-event 30-minute rest session will serve as an initial screen, not proof of a stable long-term false-command rate. Longer recordings and repeated fitting sessions will quantify variability. Ordinary arm movement and other non-command activity will be included so rejection performance is not represented only by quiet seated rest.

Transparent reporting

Results will identify the hardware revision, host device, firmware and model versions, electrode placement, and calibration procedure. This will make progress traceable across electronics, mechanical, and software iterations. Initial single-wearer results will describe that prototype configuration rather than imply equivalent performance for every user.

7 Cost, responsible development, and next stage

A low-cost hardware objective

The hardware design targets a total below \$150, using the connected phone or compatible device for classification rather than including a separate high-performance inference platform in the wearable. The current estimate is \$117.76–\$142.85, including contingency. These figures come from the September 10, 2026 prototype cost breakdown; they are estimates rather than current supplier quotations.

Prototype item	Estimated cost (USD)
nRF52840 development board	\$24.95
ADS1292R analog front end	\$17.91–\$20.00
Protected 500 mAh LiPo and disposable electrodes	\$17.90
Leads, PCB, passives, and protection	\$32.00–\$51.00
Strap, enclosure, switch, wire, and fasteners	\$15.00–\$18.00
Contingency	\$10.00
Estimated total	\$117.76–\$142.85

Totals are calculated from the listed components. Tools, the host device, shipping, and tax are excluded; the estimate does not represent a production price or include software-development costs.

Battery operation and research safety

The protected 500 mAh battery targets four-hour operation. Runtime testing will include acquisition, onboard preprocessing, and continuous Bluetooth transmission. The enclosure will retain the battery and protect the wearer from exposed electronics.

Cera One is a research prototype, not a medical device or a safety-critical controller. Body-connected operation will use only an enclosed protected battery, with USB, charging, programming, bench supplies, and grounded instruments disconnected. Reviewed electrical design and isolated bench verification will precede supervised skin-contact evaluation; a protected battery alone does not establish electrical safety.

User control and data privacy

Human testing will follow the applicable institutional review determination, accessible consent, voluntary withdrawal, and comfort-related stop conditions. A physical output-disable control will complement the host interface. Raw recordings and event logs will remain local by default, with explicit user control over recording, export, and deletion. Invalid sensing data or a lost connection will disable command output.

From prototype design to an integrated demonstration

The next stage will connect the revised sensing electronics to the host-based recognition pipeline and demonstrate the initial media-control vocabulary. Development will then extend to repeatable fitting, wireless reliability, runtime, and supervised usability evaluation. The project builds on the early PCB work and P02 wearable design, with the goal of a compact, personalized interface whose behavior is supported by documented measurements.

Project basis. Cera One prototype report, September 10, 2026; P02 wearable visualization; and the updated host-classification architecture. Performance figures are development targets.